When Does Market Definition Matter for Identification?

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Abstract

Market definition is central to demand estimation, merger simulation, and welfare analysis, yet in practice it is often chosen by researcher judgment. This paper asks when that choice matters for identification of the price coefficient. The utility parameters of aggregate logit demand can be estimated consistently without specifying the true market, provided the researcher observes a finer nested grouping and uses sufficiently fine fixed effects. This delivers a consistent benchmark against which candidate market definitions can be compared via a cluster-robust extension of the test of Papke and Wooldridge (2023) to instrumental variables. The diagnostic rejects when candidate misspecification distorts the relevant IV moment for the coefficient of interest, and remains silent when it does not. Monte Carlo evidence and an application to U.S. domestic airline demand illustrate the framework. The approach brings a foundational modeling choice within the econometric analysis rather than treating it as a maintained assumption.

[*] Views and opinions expressed reflect those of the author and do not necessarily reflect those of the United States. I thank Milena Almagro, Oren Ziv, and Jeffrey Wooldridge for helpful conversations. All mistakes are my own.

1 Introduction

Market boundaries shape every object of interest implied by a structural demand model: market shares, substitution patterns, elasticities, diversion ratios, markups, merger simulations, and welfare calculations. Yet in applied work, those boundaries are typically chosen by institutional convention or researcher judgment rather than by statistical evidence. The empirical IO literature has produced a deep set of methods for estimating demand once a market is fixed, but the specification of the market itself has remained comparatively under-disciplined. This paper asks a narrower question that nevertheless underlies most empirical demand work: when does the choice of market definition matter for identification of an endogenous price coefficient?

The answer turns out to be sharp. Market definition matters for identification when candidate misspecification distorts the relevant IV moment for the coefficient of interest, and does not matter when it leaves that moment intact. The paper develops this answer in two steps: an identification result that delivers a benchmark estimator without requiring market definition, and a fixed-effects diagnostic that compares candidate specifications against that benchmark. Applications to U.S. airline demand and bank deposit markets illustrate the framework in practice.

The identification result is for aggregate logit demand of the type studied by Berry (1994). Although correct market definition is essential for structural counterfactuals, it is not necessary for consistent estimation of the utility parameters themselves. If the researcher observes a grouping nested within the true market, then sufficiently fine 'submarket' fixed effects absorb the market-level nuisance that would otherwise require the researcher to specify the true market boundary in advance. This is useful when true market boundaries are ambiguous but a researcher has strong priors on submarkets.

The logic is visible in the standard aggregate logit system. Using the notation of Berry (1994),

$$\ln[s_{j,m}] - \ln[s_{0,m}] = \beta X_{j,m} + \alpha p_{j,m} + \delta_{j,m}, \tag{1}$$

$$\ln[q_{j,m}] = \beta X_{j,m} + \alpha p_{j,m} + \delta_{j,m} + \lambda_m, \tag{2}$$

where $s_{j,m} = q_{j,m}/Q_m$ and $\lambda_m = \ln(Q_m) + \ln(s_{0,m})$ is common to products in market m. If the researcher observes submarkets nested within m, then submarket fixed effects can absorb λ_m even when the true market boundary is not imposed directly. Given instruments Z and submarket indicators $D_{g(m)}$, the utility parameters are identified if the IV moment conditions hold conditional on X and the submarket fixed effects.

The second contribution turns this identification result into a practical diagnostic. A finer fixed-effects specification provides a consistent benchmark for the price coefficient; structural estimators based on candidate market definitions can then be compared against that benchmark. I implement the comparison using a cluster-robust extension of the Hausman test of Papke and Wooldridge (2023) (PW) to instrumental variables. The test asks whether the candidate specification is sufficient, under the maintained IV strategy, to recover the same coefficient as the fine benchmark. Equivalently, it tests whether the coarser specification leaves residual correlation between the instruments and the unobserved component of demand.

This framing produces a clean answer to the paper's central question. Monte Carlo evidence shows that the diagnostic rejects when candidate misspecification distorts the IV moment for the price coefficient and remains silent when it does not. When the instrument is uncorrelated with the omitted market-level component, even a misspecified market may deliver a similar estimate of α, and the test does not reject—which is itself useful information for the researcher. When the instrument inherits market-level dependence correlated with omitted demand variation, misspecified markets do affect the coefficient and the test rejects. The diagnostic localizes precisely when market definition matters under the researcher's identification strategy, rather than treating market correctness as a maintained assumption. Appendix B reports a separate Monte Carlo where the data-generating process is a random-coefficients logit demand model and shows that the diagnostic remains informative as a specification check even when the exact aggregate-logit algebra is no longer available.

Correct market definition remains necessary for the structural objects that motivate most applied demand work: diversion ratios, markups, merger simulation, and welfare calculations all depend on shares, the outside option, and the partition of substitutes. The framework developed here does not eliminate the need for market definition. It provides a consistent benchmark for the utility parameters in the absence of a fixed market boundary, and a way to evaluate candidate boundaries against that benchmark for the purpose of coefficient identification.

I apply the framework to U.S. domestic airline demand using the framework of Berry and Jia (2010) and the data of Aryal et al. (2024). The application compares nested market definitions ranging from airport pairs to broader city-pair markets and consistently rejects city-pair markets in favor of more disaggregated definitions, with mixed evidence for the intermediate definition. A second application to bank deposit markets, reported in Appendix D, illustrates how the test localizes the source of apparent disagreement across specifications: an initial rejection of coarser geographic markets is shown to reflect a sample-composition difference rather than market misspecification, once a common-sample restriction is imposed.

Related Literature Antitrust market definition has a substantial econometric tradition built around the SSNIP / hypothetical-monopolist test, with merger-simulation foundations (Werden and Froeb, 1994; Crooke et al., 1999) and structural extensions within discrete-choice demand models (Capps et al., 2003). Within empirical IO more broadly, market boundaries are most often disciplined informally through alternative aggregations or overidentifying restrictions. In a survey of empirical BLP applications published in top-five journals from 2018 to 2024, Zhang (2026) documents that more than 80 percent adopt ad hoc market specifications and fewer than a fifth report any robustness exercise.

Closest in spirit to this project, Zhang (2026) establishes point identification for the multiplicative scale γM_t that pins down the outside-option share in random-coefficients logit. As she emphasizes, market *size* and market *definition* are distinct objects: size influences aggregate quantities and the outside-option share, while definition partitions substitutes. Her contribution and this paper address sibling specification problems within a common methodological spirit and offer complementary tests.

The identification result here relies on the parametric structure of logit demand in a way that complements the nonparametric theory of Berry and Haile (2014). Their framework, identification via connected substitutes and an index restriction, requires observing the complete vector of market shares within each market, which presupposes that the market boundary is correctly specified. The logit functional form makes the market-level nuisance λ_m additively separable and product-invariant, so that it can be absorbed by fixed effects without observing the share vector at the true market level. The tradeoff is clear: the nonparametric approach places no restriction on the demand model but requires correct markets; the parametric approach restricts the model but relaxes the market requirement for coefficient identification.

The identification logic also has prior empirical application in urban economics; Watson and Ziv (2025, 2026) use submarket fixed effects to study housing demand in New York City, where the relevant search boundary is unknown but generally considered wider than the Census tract (Mast and Barca, 2026).

Extensions The framework extends naturally to nested logit demand, where the nest structure introduces a second dimension of specification choice alongside the market definition. Appendix A shows that the direct regression identifies the composite parameter $\theta_\alpha = \alpha/(1-\sigma)$ without requiring either the correct market or the correct nest. When nests are functions of observable categorical characteristics, the full interaction of those characteristics provides a fine benchmark that refines every candidate nesting schedule by construction. This delivers a diagnostic for nest structure, a specification choice that has historically been even more under-disciplined than market definition.[1]

The remainder of the paper proceeds as follows. Section 2 presents the aggregate logit framework and the main identification result. Section 3 develops the PW-based test for OLS and 2SLS and discusses the scope and limits of the diagnostic. Section 4 reports Monte Carlo evidence. Section 5 reports the airline application. Section 6 concludes. The appendices develop the nested-logit extension, a random-coefficients robustness simulation, additional details for the airline application, and the bank deposit application.

2 Aggregate Logit Demand, Submarkets, and Identification

This section introduces the aggregate logit model and states the paper's main identification result.

[1] An alternative more flexible approach to this problem is Almagro et al. (2025), who develop an iterative clustering algorithm that allows for arbitrary groupings of products into nests.

2.1 Aggregate Logit Setup

Let j index products, m index markets, and t index time. Assume the conditions under which aggregate demand at the market level can be written as

$$s_{jmt} = \frac{\exp(\alpha p_{jmt} + \beta X_{jmt} + \delta_{jmt})}{1 + \sum_{k \in J_m} \exp(\alpha p_{kmt} + \beta X_{kmt} + \delta_{kmt})}, \tag{3}$$

where s_{jmt} is the market share of product j, X_{jmt} are observed product characteristics, p_{jmt} is price, and δ_{jmt} is an unobserved product characteristic. Using the standard inversion argument of Berry (1994), aggregate logit demand implies

$$\ln[s_{jmt}] - \ln[s_{0mt}] = \alpha p_{jmt} + \beta X_{jmt} + \delta_{jmt}, \tag{4}$$

where s_{0mt} is the outside good share.

The standard empirical approach is to estimate equation (4) by 2SLS, possibly with fixed effects, after choosing a market definition and constructing product and outside good shares. This is where market definition enters structural demand estimation. If the market is defined incorrectly, then the implied shares, elasticities, diversion ratios, welfare calculations, and merger simulations will generally also be incorrect.

2.2 Market Definition as a Fixed-Effects Problem

A simple but important observation is that market share is a ratio of quantity sold to total market quantity, $s_{jmt} = q_{jmt}/Q_{mt}$, so that

$$\ln[s_{jmt}] = \ln[q_{jmt}] - \ln[Q_{mt}]. \tag{5}$$

Substituting this into equation (4) yields

$$\begin{aligned} \ln[q_{jmt}] &= (\ln[Q_{mt}] + \ln[s_{0mt}]) + \alpha p_{jmt} + \beta X_{jmt} + \delta_{jmt} && (6)\\ &= \lambda_{mt} + \alpha p_{jmt} + \beta X_{jmt} + \delta_{jmt} && (7)\\ &= \alpha p_{jmt} + \beta X_{jmt} + \xi_{jmt}, && (8) \end{aligned}$$

where $\lambda_{mt} \equiv \ln[Q_{mt}] + \ln[s_{0mt}]$ and $\xi_{jmt} \equiv \delta_{jmt} + \lambda_{mt}$.

Equation (8) is central to the paper. It shows that, for estimation of the utility parameters (α, β), the market-level objects $\{Q_{mt}, s_{0mt}\}$ enter only through the common nuisance term λ_{mt}. From the perspective of coefficient estimation, the main challenge is to control for the market-level component that would otherwise remain in the error term. This reframes market definition as a fixed-effects problem. If the researcher can absorb λ_{mt} using sufficiently fine fixed effects, then consistent estimation of the utility parameters does not require the true market to be specified explicitly in advance. Additionally, equation (8) clarifies the reduced-form content of α: the own-price semi-elasticity of demand.

2.3 Markets, Submarkets, and the Identification Result

Let g denote an observed grouping that is nested within the true market m. I refer to such a grouping as a *submarket*. Examples include Census tracts nested within counties, counties nested within commuting zones, or airport pairs nested within broader city-pair markets. Suppose that candidate market definitions can be ordered hierarchically as

$$g_1 \subset g_2 \subset \cdots \subset g_G.$$

If a known submarket grouping exists and the data contain sufficient within-submarket variation, then one can estimate equation (8) using submarket-by-time fixed effects rather than true market fixed effects. Let D_{gt} denote a full set of submarket-time indicators. The following result formalizes the paper's basic identification claim.

Proposition 1 (Identification without market definition). *Consider the aggregate logit demand model above. Suppose the researcher observes a grouping g such that each g-cell is contained in a single m-cell for the true market m, and suppose there exists an instrument vector Z_{jmt} satisfying*

$$\mathsf{E}[Z'_{jmt}\xi_{jmt} \mid X_{jmt}, D_{gt}] = 0, \tag{9}$$

together with the standard relevance condition for price. Then the utility parameters (α, β) are identified from the direct regression (8) using D_{gt} as fixed effects.

The proof is immediate: because $g \subset m$, the indicators D_{gt} absorb λ_{mt}, and the moment condition (9) delivers consistency under standard regularity conditions. The substantive content of Proposition 1 is that one can replace the assumption of the correct market with a potentially weaker assumption of the correct *sub*market.[2]

Correct Market Definition Still Matters Proposition 1 should not be read as implying that market definition is unimportant. Correct market definition remains necessary for the structural objects that motivate most applied demand analysis. First, elasticities depend on market shares. In the simple logit model, the own-price elasticity is $\varepsilon_{jmt} = \alpha p_{jmt}(1 - s_{jmt})$, so a misspecified market definition changes the relevant share and therefore the implied elasticity. Second, diversion ratios, markups, and merger simulation all depend on the market structure used to define substitution patterns. Third, welfare calculations depend on the outside option and total market size, which depend on market definition.

3 Testing Candidate Market Definitions

The identification result of Section 2 suggests a natural testing strategy. A nested submarket specification yields a consistent but typically less efficient estimator of the utility parameters, relative to a correctly specified structural market. Under the null that the candidate market is correct, the two estimators agree asymptotically. Under an alternative in which the candidate is too coarse in a way that matters for the relevant identifying moment, they may differ systematically. To implement this comparison, I adapt the specification test of Papke and Wooldridge (2023) (hereafter PW) to the 2SLS setting used in aggregate logit applications.

3.1 The PW Test for OLS

Papke and Wooldridge (2023) develop a robust Hausman-style test for whether a more aggregated fixed-effects specification is sufficient for estimating a coefficient of interest. The key feature of their single-coefficient test is that it avoids the nonrobust variance formula in

[2] Watson and Ziv (2025, 2026) argue that, while true market boundaries for urban housing markets are ambiguous, Census tracts are likely nested within the true market, and so they rely on within-tract variation for identification.

the classical Hausman test. Instead, it constructs an observation-level score whose dispersion estimates the sampling variance of the coefficient difference, and obtains the standard error from a constant-only regression with cluster-robust standard errors.

Let $r \in \{F, C\}$ index two fixed-effects specifications, where F denotes the finer benchmark and C denotes the coarser candidate. Let x_1 be the scalar regressor of interest and let $W^{(r)}$ collect the included controls and fixed effects under specification r. Let $\tilde{y}_i^{(r)}$ and $\tilde{x}_{1,i}^{(r)}$ denote residuals from partialling out $W^{(r)}$ from y_i and x_{1i}, respectively. The OLS coefficient under specification r is

$$\hat{\beta}^{(r)} = \frac{\sum_i \tilde{x}_{1,i}^{(r)} \tilde{y}_i^{(r)}}{\sum_i \tilde{x}_{1,i}^{2(r)}}. \tag{10}$$

Let $\hat{u}_i^{(r)} = \tilde{y}_i^{(r)} - \hat{\beta}^{(r)} \tilde{x}_{1,i}^{(r)}$ and $\hat{a}^{(r)} = n^{-1} \sum_i \tilde{x}_{1,i}^{2,(r)}$.

Under the null that the fine and candidate specifications identify the same coefficient,

$$H_0 : \beta^{(F)} - \beta^{(C)} = 0,$$

the usual asymptotic linear representation gives

$$\sqrt{n}(\hat{\beta}^{(F)} - \hat{\beta}^{(C)}) = n^{-1/2} \sum_i \left(\frac{\tilde{x}_{1,i}^{(F)} \hat{u}_i^{(F)}}{\hat{a}^{(F)}} - \frac{\tilde{x}_{1,i}^{(C)} \hat{u}_i^{(C)}}{\hat{a}^{(C)}} \right) + o_p(1). \tag{11}$$

Define the observation-level score as

$$\hat{q}_i^{OLS} = \frac{\tilde{x}_{1,i}^{(F)} \hat{u}_i^{(F)}}{\hat{a}^{(F)}} - \frac{\tilde{x}_{1,i}^{(C)} \hat{u}_i^{(C)}}{\hat{a}^{(C)}}. \tag{12}$$

Because each component of $\hat{q}_i^{OLS}$ satisfies its own OLS first-order condition, its sample mean is zero. Following PW, one can regress $\hat{q}_i^{OLS}$ on a constant, and obtain a robust standard error for the difference in coefficients.

Thus, the PW test statistic is

$$t_{PW} = \frac{\hat{\beta}^{(F)} - \hat{\beta}^{(C)}}{\widehat{se}(\hat{\beta}^{(F)} - \hat{\beta}^{(C)})}, \tag{13}$$

where $\widehat{se}(\hat{\beta}^{(F)} - \hat{\beta}^{(C)})$ is the robust standard error from the constant-only regression of $\hat{q}_i^{OLS}$

on an intercept. The clustering level should allow for the dependence structure induced by the more aggregated specification; in the applications below I cluster at the coarser market level.

3.2 PW Adapted to 2SLS

Aggregate logit demand is typically estimated by instrumental variables, so the diagnostic must account for the first-stage projection. I use the same PW logic, replacing the residualized regressor in the OLS influence representation with the fitted value from the residualized first stage.

For specification $r \in \{F, C\}$, let $W^{(r)}$ collect the included controls and fixed effects, and let $\tilde{y}^{(r)} = M^{(r)}y$, $\tilde{x}^{(r)} = M^{(r)}x$, and $\tilde{Z}^{(r)} = M^{(r)}Z$ denote residualized outcome, endogenous regressor, and instruments, respectively. The first-stage fitted value is $\hat{x}^{(r)} = P_{\tilde{Z}^{(r)}}\tilde{x}^{(r)}$, and so the scalar 2SLS estimator can be written as

$$\hat{\beta}_{IV}^{(r)} = \frac{\sum_i \hat{x}_i^{(r)} \tilde{y}_i^{(r)}}{\sum_i \hat{x}_i^{(r)} \tilde{x}_i^{(r)}}. \tag{14}$$

Let $\hat{u}_i^{(r)} = \tilde{y}_i^{(r)} - \hat{\beta}_{IV}^{(r)} \tilde{x}_i^{(r)}$ and $\hat{a}^{(r)} = n^{-1} \sum_i \hat{x}_i^{(r)} \tilde{x}_i^{(r)}$.

The 2SLS analogue of equation 12 is then

$$\hat{q}_i^{2SLS} = \frac{\hat{x}_i^{(F)} \hat{u}_i^{(F)}}{\hat{a}^{(F)}} - \frac{\hat{x}_i^{(C)} \hat{u}_i^{(C)}}{\hat{a}^{(C)}}, \tag{15}$$

and similarly the resulting test statistic is

$$t_{PW\text{-}2SLS} = \frac{\hat{\beta}_{IV}^{(F)} - \hat{\beta}_{IV}^{(C)}}{\widehat{se}(\hat{\beta}_{IV}^{(F)} - \hat{\beta}_{IV}^{(C)})}, \tag{16}$$

where the denominator is obtained from the cluster-robust standard error in the constant-only regression of $\hat{q}_i^{2SLS}$ on an intercept.

The interpretation for equation 16 is the same as the OLS result in PW. The test does not ask whether the coarser fixed effects are the true structural fixed effects. Rather, it asks whether using the candidate fixed-effects structure changes the identified coefficient relative to the finer benchmark, under the maintained IV strategy. In the market-definition setting, rejection means that the candidate market definition is not adequate for recovering the same price coefficient as the fine fixed-effects benchmark.

3.3 Application to Aggregate Logit Market Definitions

In the aggregate logit setting, the test evaluates whether a coarser market-level fixed-effects structure recovers the same price coefficient as a finer benchmark. Let the fine specification be the direct log-quantity regression in equation (8) with submarket-by-time fixed effects, and let the candidate specification use a coarser market-by-time fixed effect. Under the null that the candidate fixed effects are sufficient for the maintained IV moment, both estimators identify the same α and the PW-2SLS difference is asymptotically zero. Under an alternative in which the candidate market is too coarse in a way that leaves residual correlation between the instruments and the unobserved demand component, the two estimators diverge.

3.4 Scope and Limits of the Diagnostic

Three caveats clarify the interpretation of the test and motivate the simulation design in the next section. First, the test concerns whether the chosen specification is sufficient for the relevant orthogonality condition, not whether that specification recovers the entire structural model. The price coefficient is the natural target because it is the parameter most often used in counterfactual exercises, but the same construction can be applied to any coefficient identified from the model.

Second, the test is informative only when misspecification affects the moment condition for the coefficient of interest. If the parameter of interest is the price coefficient and the instrument is uncorrelated with the omitted market-level component, then even a misspecified market definition may deliver a similar estimate of α, and the test will have limited power. This is not a defect: the test is precisely diagnosing whether market misspecification matters under the researcher's identification strategy. A non-rejection in this case is informative in its own right: it tells the researcher that the choice of market boundary is unlikely to affect the coefficient of interest.

Third, the test may conflate market misspecification with the adequacy of the maintained IV strategy. If the instrument fails the exclusion restriction at the candidate's level of aggregation but holds after conditioning on a finer fixed-effects partition, the candidate and fine estimators will diverge whether or not the candidate is the true market. The test therefore does not directly adjudicate market correctness. Rather, under the researcher's maintained IV strategy, it asks whether the candidate specification is sufficient to recover the same coefficient as the fine benchmark. The test should be read as a diagnostic that disciplines a modeling choice, not as a verdict on the structural model.

3.5 Nested Logit Extension.

The framework also extends to nested logit demand. Product nests are a second dimension of specification choice that arises alongside the market definition. The Berry inversion now introduces the within-nest share $\ln(s_{j|g,m})$ as an additional endogenous regressor and a nesting parameter σ governing within-nest substitutability. The direct regression analogue identifies only a composite parameter $\theta_\alpha \equiv \alpha/(1-\sigma)$, but the nuisance term Λ_{gm} remains constant within nest-by-market cells, so sufficiently fine fixed effects can again absorb it without specifying either the true nests or the true markets. When nests are functions of observable categorical characteristics, the full interaction of those characteristics provides a universal fine benchmark that refines every candidate nesting schedule by construction. Appendix A develops the full extension; the nest-testing case is empirically more demanding because fixed-effects cells multiply across two dimensions.

4 Monte Carlo Evidence

This section reports a Monte Carlo exercise designed to illustrate when the PW comparison is informative for market definition in aggregate logit demand. The simulation distinguishes cases in which the excluded instrument is uncorrelated with market-level omitted demand variation from cases in which the instrument inherits such dependence. The central message is that the test's behavior tracks the substantive importance of market misspecification for the coefficient of interest: it is most informative precisely when getting the market right matters.

I use the language of 'states,' 'counties,' and 'tracts' for concreteness, where counties are the true market. The structure consists of 50 states, 10 counties per state, and 5 tracts per county, for a total of 500 true markets and 2, 500 submarkets. The true price coefficient is $\alpha = -1.5$. I conduct 500 replications. I simulate the data using `pyblp` and their solver for endogenous prices.

4.1 Data-Generating Process

The data-generating process is an aggregate-logit model with multi-product firms. Let m denote the true market and let $g \subset m$ denote a strictly nested subgroup. Indirect utility is

$$u_{ijm} = \beta_0 + \alpha p_{jm} + \beta_1 x_{1,jm} + \beta_2 x_{2,jm} + \xi_{jm} + \varepsilon_{ijm}, \tag{17}$$

where ξ_{jm} is a product-level demand shock and ε_{ijm} is an idiosyncratic Type I extreme-value error. I also specify a marginal cost function, that is linear in (x_1, x_2, Z). Equilibrium prices are obtained from the multi-product Bertrand first-order conditions. The excluded instrument is

$$Z_{jm} = z^p_{jm} + z^m_m, \tag{18}$$

where z^p_{jm} is a purely idiosyncratic product-level component and z^m_m is a market-level component. The simulation scenarios vary in whether z^m_m is independent of ξ_{jm} or correlated with it, and in whether the candidate structural market coincides with the true market or is too coarse or too small. Market-level dependence in ξ_{jm} is modeled as $\xi_{jm} = \xi^p_{jm} + \eta_m$, where ξ^p_{jm} is purely idiosyncratic. Quantity is $q_{jm} = M s_{jm}$, where M is market size.

4.2 Estimators

The four estimators correspond directly to the paper's theoretical discussion. The first is the candidate structural model, which may use the correct market definition or one that is too coarse or too small. The remaining three are direct specifications, each using one tier of the potential market definitions: state, county, or tract. For candidate market m and tiers s, c, t:

$$\text{Structural}: \ \log(s_{jm}) - \log(s_{0m}) = \alpha p_{jm} + \beta_1 x_{1,jm} + \beta_2 x_{2,jm} + u_{jm}, \tag{19}$$

$$\text{Direct-State}: \ \log(q_{jtcs}) = \alpha p_{jtcs} + \beta_1 x_{1,jtcs} + \beta_2 x_{2,jtcs} + \pi_s D_{s(j)} + u_{jtcs}, \tag{20}$$

$$\text{Direct-County}: \ \log(q_{jtcs}) = \alpha p_{jtcs} + \beta_1 x_{1,jtcs} + \beta_2 x_{2,jtcs} + \pi_c D_{c(j)} + u_{jtcs}, \tag{21}$$

$$\text{Direct-Tract}: \ \log(q_{jtcs}) = \alpha p_{jtcs} + \beta_1 x_{1,jtcs} + \beta_2 x_{2,jtcs} + \pi_t D_{t(j)} + u_{jtcs}, \tag{22}$$

where s_{0m} is the outside share for the candidate market and D_τ are indicator variables for nesting tier τ. All estimators use Z_{jm} as the excluded instrument. When the candidate market is correct, equation (19) is the properly specified structural regression; when it is too coarse or too small, the inversion uses shares and outside shares constructed at the wrong level.

4.3 Simulation Scenarios

The simulation varies along two dimensions. First, the candidate structural market may be correct, too coarse, or too small relative to the true market. Second, the excluded instrument may be orthogonal to the omitted market-level demand component or inherit market-level dependence correlated with it. Together these choices yield the six scenarios reported in Tables 1 and 2.

The baseline scenario uses the correct candidate market and a clean instrument; all four estimators are consistent in this design and the PW comparisons should not systematically reject. The second scenario again uses the correct candidate market but with a contaminated instrument: now the structural and Direct-State estimators no longer satisfy the IV moment condition, while the Direct-County and Direct-Tract specifications remain valid because they absorb the omitted county-level component. The third and fourth scenarios consider a too-coarse candidate market, with and without instrument contamination. The fifth and sixth do the same for a too-small candidate. These six cases jointly span the empirically interesting combinations of misspecification (in the structural market) and instrument behavior.

4.4 PW Comparisons

For each scenario, I compute four PW-2SLS tests: (1) County vs. State, (2) Tract vs. State, (3) Tract vs. County, and (4) Tract vs. Structural. The Direct-Tract and Direct-County estimators are consistent throughout. When the instrument is uncorrelated with market conditions and the structural market is correctly specified, the structural specification is also consistent. All other specifications will be inconsistent.

4.5 Results: When Does Market Definition Matter?

The results provide a sharp characterization of when market definition matters for the price coefficient under the maintained identification strategy.

Coefficient estimates. Table 1 reports Monte Carlo means and standard deviations of the estimated price coefficient across scenarios. Three features stand out. First, the Direct-County and Direct-Tract specifications are tightly centered around the true value in every scenario. This is exactly the role they are designed to play: because counties are always the true market, county fixed effects absorb the omitted county-level component and ensure instrument validity, while tract fixed effects provide an even finer benchmark that remains consistent. Second, the structural candidate approach performs substantially worse whenever it is misspecified with extreme coefficient volatility. Even when the instrument is uncorrelated with market conditions, the structural estimate can be strongly biased downward; with the contaminated instruments the coefficient is less volatile. Third, the Direct-State estimates are attenuated toward zero whether the instruments are clean or contaminated, though the mechanism differs across the two cases. With contaminated instruments, Z inherits county-level dependence and the IV moment fails directly. With clean instruments, the IV moment for Z is satisfied, but state fixed effects do not absorb the county-level demand component η_m, which remains in the residual.

PW comparisons. Table 2 reports the PW comparisons. The Tract-vs-County comparison rarely rejects in any scenario, with rejection rates close to nominal size and mean t-statistics near zero. This is consistent with the role of tract fixed effects as a finer but less efficient benchmark relative to the correctly specified county direct regression: both

Table 1 – Monte Carlo Coefficient Results

	Structural Candidate (1)	State (2)	Direct County (3)	Tract (4)
Correct candidate, $Z \perp \eta_c$	-1.50	-1.40	-1.50	-1.50
	(0.07)	(0.06)	(0.01)	(0.01)
Correct candidate, $Z \not\perp \eta_c$	-0.42	-0.43	-1.50	-1.50
	(0.04)	(0.03)	(0.01)	(0.01)
Too-coarse candidate, $Z \perp \eta_c$	-4.15	-1.40	-1.50	-1.50
	(0.91)	(0.06)	(0.01)	(0.01)
Too-coarse candidate, $Z \not\perp \eta_c$	-0.56	-0.43	-1.50	-1.50
	(0.31)	(0.04)	(0.01)	(0.01)
Too-small candidate, $Z \perp \eta_c$	-1.43	-1.41	-1.50	-1.50
	(0.06)	(0.06)	(0.01)	(0.01)
Too-small candidate, $Z \not\perp \eta_c$	-0.41	-0.43	-1.50	-1.50
	(0.04)	(0.04)	(0.01)	(0.01)

Note: Each first line reports Monte Carlo means for the estimated price coefficient across 500 iterations. The line below reports Monte Carlo standard deviations in parentheses. The true market is county in every scenario; the Structural Candidate column uses shares and outside shares constructed at the candidate market level, while the Direct columns estimate $\log(q)$ regressions with state, county, or tract fixed effects.

estimators are consistent, so the test should not systematically reject.

The remaining comparisons identify two distinct mechanisms by which market definition can matter for $\hat{\alpha}$. When the instrument is orthogonal to the omitted county-level component, the IV moment is satisfied at every level of aggregation, but state fixed effects still fail to absorb η_m. The County-vs-State and Tract-vs-State comparisons reject in roughly 30-37% of replications in this design, reflecting finite-sample power against a real difference in probability limits; rejection rates would likely rise toward unity in larger samples. When the instrument inherits county-level dependence, the IV moment additionally fails for these specifications and the same comparisons reject in essentially every replication. The Tract-vs-Structural comparison rejects when the structural candidate is misspecified, either because the candidate market is wrong, the IV moment fails, or both. Under the correct candidate with a clean instrument, this comparison stays near nominal size.

Taken together, the simulation supports interpreting the PW-2SLS comparison as a diagnostic for whether the candidate specification yields a different probability limit for α than the fine benchmark. A non-rejection is informative in its own right: it tells the researcher that, under the maintained IV strategy and the available fixed-effects partition, the choice of market boundary is unlikely to drive the price coefficient.

Table 2 – PW Comparisons in the Monte Carlo

	$Z \perp \eta_c$		$Z \not\perp \eta_c$	
	Mean t-statistic	Rejection rate	Mean t-statistic	Rejection rate
Correct, County-State	-1.62	0.37	-30.77	1.00
Correct, Tract-State	-1.61	0.36	-30.18	1.00
Correct, Tract-County	-0.02	0.03	-0.01	0.06
Correct, Tract-Struct.	-0.05	0.04	-29.94	1.00
Too-coarse, County-State	-1.59	0.34	-30.70	1.00
Too-coarse, Tract-State	-1.58	0.34	-30.17	1.00
Too-coarse, Tract-County	0.01	0.06	-0.03	0.05
Too-coarse, Tract-Struct.	2.96	0.84	-8.69	0.79
Too-small, County-State	-1.48	0.30	-30.47	1.00
Too-small, Tract-State	-1.47	0.31	-29.92	1.00
Too-small, Tract-County	-0.01	0.05	-0.00	0.05
Too-small, Tract-Struct.	-1.19	0.22	-31.38	1.00

Note: The table reports PW comparisons based on 500 Monte Carlo replications. Rejection rate is the fraction of replications in which the two-sided PW test rejects at the 5 percent level.

4.6 Extension to Random Coefficients Logit Demand

Modern demand applications often allow richer substitution patterns than aggregate logit. In Appendix B, I conduct a separate Monte Carlo where the data generating process matches a 'BLP-style' random coefficients logit demand structure. The exact fixed-effects identification result in the paper is specific to aggregate logit and nested logit, so this exercise is best interpreted as a robustness and external-validity check rather than as a new theorem. Overall, the results suggest that the PW diagnostic remains as informative as a specification diagnostic even when the exact aggregate-logit algebra is no longer available.

5 Application: Air Travel Demand

I apply the diagnostic to U.S. domestic airline demand using the replication data from Aryal et al. (2024), originally drawn from the DOT DB1B Airline Origin and Destination Survey. This setting is well-suited to the exercise because the candidate market definitions are naturally hierarchical: airport pairs are nested within origin-airport by destination-city pairs, which are in turn nested within city pairs. The application asks whether coarser market definitions are adequate for estimating the price-income coefficient under the maintained IV strategy.

5.1 Data and Market Hierarchy

The data come from the second quarter of each year from 1998 to 2016. The DB1B is a 10% sample of airline tickets, and I follow standard practice in scaling passenger counts by ten to recover estimated totals. Market size is defined as $M_m = 0.9 \times \sqrt{\text{pop}_o \times \text{pop}_d}$, where origin and destination populations are 2010 Census CBSA populations. The outside share is computed as $s_{0m} = 1 - \sum_j s_{jm}$ at each candidate market definition. After standard sample restrictions, the common estimation sample contains 863,182 carrier-market-year observations across 18,982 airport pairs and 16 carriers.

I consider three nested market definitions. The finest is a directional airport pair, such as DCA–ORD. The intermediate market is origin-airport by destination-city, so that DCA–Chicago pools ORD and MDW at the destination. The coarsest is the city pair, such as Washington–Chicago. Airport-to-MSA assignments are based on a static IATA-to-CBSA crosswalk. By construction, each finer definition is nested within the next coarser one, which is the structure required for the PW test.[3]

5.2 Empirical Setup

I estimate the price-income coefficient under two approaches. The first is the direct estimator implied by Section 2.2, which serves as the finer fixed-effects benchmark in the testing framework:

$$\ln[q_{jmt}] = \alpha(p_{jmt}/y_{ot}) + \beta X_{jmt} + \mu_j + \lambda_{mt} + u_{jmt}, \tag{23}$$

[3] Airport-pairs are not *strictly* nested, because some cities only have a single airport; e.g., if two cities only have a single airport then the airport pair is identical to the city pair.

where the market definition enters through interacted market-by-year fixed effects. The second is the standard structural logit estimator,

$$\ln[s_{jmt}] - \ln[s_{0mt}] = \alpha(p_{jmt}/y_{ot}) + \beta X_{jmt} + \mu_j + \lambda_m + \lambda_t + \xi_{jmt}, \tag{24}$$

where shares are recomputed under each candidate market definition and fixed effects enter separately, which is likely what an applied researcher would do in this setting.

In both specifications, the endogenous regressor is the carrier's average fare normalized by origin-market income, p_{jmt}/y_{ot}. The included controls are the carrier's nonstop passenger share and mean routing circuity. I report results for three excluded-instrument sets: own cost shifters, BLP-style rival characteristics, and the combination of the two. Construction details are reported in Appendix C. Table 9 shows that first-stage strength is comfortably above conventional thresholds in all specifications.

5.3 Results

Table 3 reports the second-stage estimates. Three patterns stand out. First, the own cost-shifter IVs yield stable and economically sensible negative coefficients across market definitions in both the direct and structural specifications. Second, in the direct specification, the BLP-rival and combined instrument sets become more negative as the market definition becomes coarser. Third, in the structural logit, the BLP-rival and combined instrument sets yield positive coefficients despite strong first stages. Together, these patterns suggest that the identifying variation in the BLP-based instruments is sensitive to the fixed-effects structure, and that the coarser structural specifications may not absorb the relevant demand-side variation.

Table 4 reports the PW tests for the direct IV estimator. The main pattern is that the city-pair and intermediate definitions are rejected in all comparisons. Only one comparison fails to reject at the 5% level: the own-cost-shifter specification comparing the fine and intermediate definitions ($p = 0.34$). Under the maintained IV strategy, these results are most consistent with airport-pair markets as the finest adequate definition in this application, while suggesting that the intermediate definition may be adequate when identification relies only on the cost-shifter instruments.

Table 3 – Price/Income Coefficient ($\hat{\alpha}$)

Market definition	Airport Pair (1)	Orig.Airport–Dest.City (2)	City Pair (3)
Panel A: Direct, Interacted market-by-year FE			
Own cost shifters	−3.19	−3.20	−3.23
	(0.14)	(0.16)	(0.16)
BLP rival chars	−1.63	−2.55	−2.63
	(0.09)	(0.15)	(0.14)
Cost + BLP	−1.44	−2.21	−2.33
	(0.07)	(0.09)	(0.09)
Market × year FE	Yes	Yes	Yes
Carrier FE	Yes	Yes	Yes
N	863,182	863,182	863,182
Panel B: Structural Logit, Separate market + year FEs			
Own cost shifters	−2.59	−2.58	−2.63
	(0.13)	(0.13)	(0.14)
BLP rival chars	1.02	1.03	0.92
	(0.03)	(0.03)	(0.03)
Cost + BLP	0.57	0.60	0.51
	(0.02)	(0.02)	(0.02)
Market FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Carrier FE	Yes	Yes	Yes
N	863,209	794,987	730,308

Note: This table reports the price/income coefficient $\hat{\alpha}$ from direct and structural logit specifications across three nested market definitions. Panel A reports the direct approach, in which the market definition governs the granularity of the interacted market-by-year fixed effects. Panel B reports the structural logit approach, in which the market definition governs both the level at which shares are computed and the market fixed effects. All specifications include carrier fixed effects and two carrier-route-year controls. Cluster-robust standard errors are clustered at the market level.

Table 4 – Papke–Wooldridge Market Definition Tests

IV Set	Comparison	$\hat{\alpha}_{\text{diff}}$	SE	p	Rej.
Own cost shifters	Fine v. Interm.	−0.01	0.015	0.343	N
	Fine v. Coarse	−0.05	0.023	0.036	Y
	Interm. v. Coarse	−0.03	0.015	0.023	Y
BLP rival chars	Fine v. Interm.	−0.91	0.048	0.000	Y
	Fine v. Coarse	−1.00	0.063	0.000	Y
	Interm. v. Coarse	−0.09	0.036	0.017	Y
Cost + BLP	Fine v. Interm.	−0.77	0.037	0.000	Y
	Fine v. Coarse	−0.90	0.049	0.000	Y
	Interm. v. Coarse	−0.13	0.028	0.000	Y

Note: This table reports Papke–Wooldridge tests comparing $\hat{\alpha}$ across pairs of nested market definitions: airport pair (fine), origin-airport × destination-city (intermediate), and city pair (coarse). Standard errors are cluster-robust at the coarser market level. 'Y' denotes rejection at the 5% level.

To display how market definition can affect empirical findings, for each market definition candidate I calculate own-price elasticity, multi-product markup, diversion to outside good, and diversion to rival carriers. For this comparison, I use a subsample where the origin or destination city has at least two airports (else, there is no difference between the airport-pair and city-pair). Table 5 reports these objects across the three market definitions, holding $\hat{\alpha}$ fixed at the airport-pair direct-IV estimate of −3.19. Fixing $\hat{\alpha}$ isolates the channel of interest: any differences across columns are generated entirely by the change in market definition rather than by re-estimation of demand curvature.[4]

In this city-pair-binding subset, moving from airport-pair to city-pair markets lowers median diversion to the outside good from 0.99 to 0.95 and raises median diversion to other carriers from 0.01 to 0.05. The corresponding median absolute product-level changes relative to the airport-pair benchmark are 0.03 for outside diversion and 0.02 for other-carrier diversion. These shifts in substitution patterns are generated entirely by the redefinition of the market: shares and the partition of inside goods change, while preferences are held fixed. Even when coefficient identification is disciplined by fine fixed effects, the structural objects that motivate most applied demand work depend on how the market is defined.

Substantively, the results suggest that travelers do not treat airports serving the same

[4]Note, diversion ratios in simple logit depend on shares and the market partition but not on $\hat{\alpha}$, so the diversion rows would be identical by construction under either parameter estimates.

Table 5 – Airline Structural Objects in City-Pair-Binding Markets

Statistic	Airport Pair	Orig.Airport–Dest.City	City Pair
Median own-price elasticity	−13.16	−13.19	−13.25
Median multi-product markup	17.69	17.69	17.72
Median diversion to outside good	0.988	0.973	0.948
Median diversion to other carriers	0.012	0.025	0.046
N	270,330	268,543	265,244
Market-years	66,034	45,521	26,177

Note: The sample is restricted to market-years where the city-pair definition pools more than one airport pair; city-pair aggregation binds for 31.3% of the common-sample observations. $\hat{\alpha}$ is fixed at the airport-pair direct-IV own-cost-shifter estimate, −3.19, so all variation across columns reflects changes in shares and the partition of inside goods rather than re-estimation of $\hat{\alpha}$.

metropolitan area as interchangeable enough for city-pair aggregation to preserve the estimated price sensitivity. More generally, the application illustrates the paper's central point: candidate market definitions can be evaluated by comparing them to a finer fixed-effects benchmark rather than imposed purely on institutional grounds.

6 Conclusion

This paper asks when the choice of market definition matters for identification of the price coefficient in aggregate logit demand. The answer is that it matters when candidate misspecification distorts the IV moment for the coefficient of interest, and does not matter when it leaves that moment intact.

The argument has two pieces. First, an identification result: correct market definition, while essential for structural counterfactuals, is not necessary for consistent estimation of the utility parameters themselves. By rewriting the model as a regression of log quantities on prices with a market-level nuisance component, I show that sufficiently fine nested fixed effects absorb the market-level term that would otherwise require the researcher to specify the true market boundary in advance. The boundary itself need not be known so long as a finer hierarchical grouping is observed and the IV moment holds conditional on the corresponding fixed effects.

Second, a diagnostic: a finer fixed-effects specification provides a consistent benchmark for the price coefficient, and structural estimators based on candidate market definitions can be compared against that benchmark using a cluster-robust extension of the

Hausman test of Papke and Wooldridge (2023) to instrumental variables. The test asks whether the candidate specification is sufficient, under the maintained IV strategy, to recover the same coefficient as the fine benchmark. Monte Carlo evidence and an application to U.S. domestic airline demand show that the diagnostic rejects when candidate misspecification distorts the IV moment for the price coefficient and remains silent when it does not. A non-rejection indicates that the candidate is adequate for the coefficient under the maintained IV strategy, not a verdict on the candidate as the true market.

The framework brings a foundational modeling choice into formal econometric scrutiny rather than treating it as a maintained assumption, and it extends naturally to the nest-structure choice in nested logit demand.

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A Extension to Nested Logit Demand

This appendix extends the paper's identification and testing framework to the nested logit model of Berry (1994). The extension introduces a second dimension of specification choice—the nest structure—alongside the market definition. I show that the direct regression identifies a composite parameter $\theta_\alpha \equiv \alpha/(1-\sigma)$ without requiring either the correct market definition or the correct nest structure, and that the PW-2SLS test can be applied to evaluate candidate specifications along either or both dimensions.

Relation to data-driven nesting. Recent independent work by Almagro et al. (2025) studies a complementary problem: how to recover the nesting structure itself in a nested-logit demand model. Their approach uses aggregate market shares and product characteristics to estimate latent nests, applying a group fixed-effect or regression-based clustering procedure in a first step and then estimating the nested-logit preference parameters as if the recovered groups were known. This is closely related in spirit to the present appendix because both approaches exploit the fact that nested-logit choice probabilities can be written so that nest-specific demand shifters appear as group-by-market components.

The objects of interest, however, are different. Their procedure is constructive: it seeks to estimate the relevant product nests. This paper is diagnostic: taking candidate nests and candidate geographic markets as given, we ask whether alternative fixed-effect structures imply asymptotically equivalent estimates of the composite price coefficient. Thus, our proposed tests should be viewed as complementary to data-driven nest recovery methods rather than as substitutes for them. In particular, data-driven methods can be used to propose economically meaningful candidate nests, while the fixed-effect comparisons developed here provide a way to test whether those candidate nests, possibly jointly with candidate market definitions, are sufficiently rich for the moment conditions underlying the demand estimates.

A.1 Nested Logit Setup

Maintain the notation of Section 2, and let g index nests within market m. The nested logit model adds a within-nest correlation parameter $\sigma \in [0, 1)$ to the utility specification, so that the Berry inversion yields

$$\ln(s_{jm}) - \ln(s_{0m}) = \beta X_{jm} + \alpha p_{jm} + \sigma \ln(s_{j|g,m}) + \delta_{jm}, \tag{25}$$

where $s_{j|g,m} \equiv s_{jm}/s_{gm}$ is the within-nest share of product j and $s_{gm} \equiv \sum_{k \in g} s_{km}$ is the nest share. In the structural specification, estimation of equation (25) by 2SLS requires correct specification of both the market m (which determines s_{jm}, s_{0m}, and s_{gm}) and the nest g (which determines $s_{j|g,m}$ and s_{gm}).

A.2 Deriving the Composite Parameter

Substitute $\ln(s_{jm}) = \ln(q_{jm}) - \ln(M_m)$ and $\ln(s_{j|g,m}) = \ln(q_{jm}) - \ln(M_m) - \ln(s_{gm})$ into equation (25) to obtain

$$\ln(q_{jm}) - \ln(M_m) - \ln(s_{0m}) = \beta X_{jm} + \alpha p_{jm} + \sigma\left[\ln(q_{jm}) - \ln(M_m) - \ln(s_{gm})\right] + \delta_{jm}. \tag{26}$$

Collecting $\ln(q_{jm})$ on the left-hand side and dividing by $(1-\sigma)$ yields

$$\ln(q_{jm}) = \underbrace{\frac{\alpha}{1-\sigma}}_{\equiv \theta_\alpha} p_{jm} + \underbrace{\frac{\beta}{1-\sigma}}_{\equiv \theta_\beta} X_{jm} + \Lambda_{gm} + \frac{\delta_{jm}}{1-\sigma}, \tag{27}$$

where the nuisance term is

$$\Lambda_{gm} \equiv \ln(M_m) + \frac{1}{1-\sigma}\ln(s_{0m}) - \frac{\sigma}{1-\sigma}\ln(s_{gm}). \tag{28}$$

Two features of equation (27) are central. First, Λ_{gm} is constant across products j within a nest-by-market cell (g, m). It depends on market-level objects (M_m, s_{0m}) and nest-by-market objects (s_{gm}), but not on the identity of product j. Second, the utility parameters α and σ enter only through the composite $\theta_\alpha = \alpha/(1-\sigma)$, and similarly for β. The composite parameters are all that can be identified from the direct regression; separate identification of α and σ requires the correctly specified structural model.

A.3 Identification via Fine Fixed Effects

The nuisance term Λ_{gm} can be absorbed by sufficiently fine fixed effects in both the nest and market dimensions. Let h denote an observed grouping satisfying $h \subset g$, and let s denote an observed submarket grouping satisfying $s \subset m$. Since each (h, s) cell is contained within a single (g, m) cell, a full set of $(h \times s)$ indicators absorbs Λ_{gm}.

Let $D_{h\times s}$ denote the corresponding indicators. Consistent estimation of θ_α requires

$$\mathsf{E}\left[Z'_{jm} \frac{\delta_{jm}}{1-\sigma} \,\middle|\, X_{jm},\, D_{h\times s} \right] = 0, \tag{29}$$

together with the standard relevance condition. Crucially, this moment condition requires neither the true market boundary m nor the true nest structure g to be specified explicitly. The researcher needs only that h is finer than g and s is finer than m.

When $\sigma = 0$, the nested logit collapses to the plain logit, $\theta_\alpha = \alpha$, the nest dimension is trivial, and the identification result reduces to Proposition 1 in the main text.

A.4 Observable Characteristics as a Universal Fine Benchmark

A natural question is where the subnest grouping h comes from. In the market dimension, geography provides a natural hierarchy: tracts nest within counties, counties within MSAs, and so on. No analogous hierarchy exists for nests, which are defined by patterns of consumer substitution rather than by an external ordering.

A simple observation resolves this problem when nests are determined by observable product characteristics. Let X^d denote the vector of categorical (discrete) product attributes, and suppose that the true nest assignment is a function of observables:

$$g_j = H(X_j^d) \tag{30}$$

for some unknown function $H(\cdot)$. Many applied nested logit specifications satisfy this condition: in airline demand, nests might be determined by stop type and carrier class; in automobile demand, by vehicle segment; in cereal, by product category.

Under equation (30), the full interaction of all elements of X^d—call it X^d_{full}—is automatically at least as fine as any function $H(X^d)$. Every value of H maps each unique X^d_{full} cell to a single nest, so X^d_{full} refines every candidate nesting schedule by construction. This means that $(X^d_{\text{full}} \times s)$ fixed effects absorb Λ_{gm} regardless of which H governs the true nest structure.

This observation has two consequences for the PW testing framework.

First, it provides a concrete fine benchmark for the "known markets, unknown nests" case described below. The researcher estimates the direct regression (27) with $(X^d_{\text{full}} \times m)$ fixed effects to obtain a consistent benchmark for θ_α, then compares against a candidate nesting schedule $\hat{g} = \hat{H}(X^d)$ using $(\hat{g} \times m)$ fixed effects. Because X^d_{full} refines every function of X^d, the fine specification is valid for any candidate $\hat{H}$, and the researcher need not know H in advance.

Second, because the fine benchmark is invariant to the candidate, one can compare multiple candidate nesting schedules $\hat{H}_1, \hat{H}_2, \ldots$ against the same benchmark and select the coarsest grouping that is not rejected by the PW test. In principle, one could also evaluate a proposed $\hat{H}^*$ against a distribution of PW statistics obtained under random re-assignment of X^d_{full} cells to nests, providing a permutation-style diagnostic for whether the proposed grouping captures genuine substitution patterns beyond what arbitrary partitions of observable space would achieve.

The practical limitation is within-cell variation. If X^d has many categorical dimensions, the full interaction can produce cells with very few products per market, leaving insufficient variation in price and instruments to identify θ_α precisely. The fine benchmark need not literally be X^d_{full}: any grouping known to be finer than the candidates under consideration is sufficient. In many applications, the substantive debate involves two or three candidate nesting schedules, and a grouping one level finer than the finest candidate will serve as an adequate benchmark without exhausting degrees of freedom.

A.5 PW Tests for Nested Logit Specifications

The PW-2SLS testing framework of Section 3 extends directly to the nested logit setting. In every case, the object being tested is θ_α, and the implementation is identical: compute the influence function under each fixed-effects specification, difference the two, and test whether the mean difference is zero using a cluster-robust regression on a constant. Three cases are of practical interest.

A.5.1 Simultaneous Test of Markets and Nests

Suppose the researcher has candidate nests $\hat{g}$ and candidate markets $\hat{m}$, together with finer groupings $h \subset g$ and $s \subset m$. Let

$$\hat{\theta}^{\text{fine}}: \quad (h \times s) \text{ fixed effects applied to equation (27)}, \tag{31}$$

$$\hat{\theta}^{\text{cand}}: \quad (\hat{g} \times \hat{m}) \text{ fixed effects applied to equation (27)}. \tag{32}$$

Under H_0, each $(\hat{g} \times \hat{m})$ cell is contained within some $(g \times m)$ cell, so both estimators are consistent for θ_α and the PW difference should vanish asymptotically. Under H_1, the candidate specification is too coarse in at least one dimension: some $(\hat{g} \times \hat{m})$ cells span multiple (g, m) cells, and Λ_{gm} is not fully absorbed.

Rejection of H_0 indicates that the joint candidate $(\hat{g}, \hat{m})$ is inadequate for θ_α, but it does not identify which dimension—nests, markets, or both—is responsible for the discrepancy.

A.5.2 Known Nests, Unknown Markets

Suppose the researcher is confident in the nest structure g (e.g., nests are determined by observable product categories and are not controversial) but uncertain about the market definition. Let $s \subset m$ be an observed submarket. The two specifications are

$$\hat{\theta}^{\text{fine}}: \quad (g \times s) \text{ fixed effects}, \tag{33}$$

$$\hat{\theta}^{\text{cand}}: \quad (g \times \hat{m}) \text{ fixed effects}. \tag{34}$$

Under H_0, $\hat{m}$ is at least as fine as the true market m, so $(g \times \hat{m})$ absorbs Λ_{gm} and both estimators are consistent for θ_α. Under H_1, $\hat{m}$ is too coarse and the market-level components of Λ_{gm} are not fully absorbed.

This is the direct analogue of the plain logit market-definition test developed in the main text, applied within the nested logit framework. The only difference is that the test concerns the composite parameter θ_α rather than α itself, and the fixed effects in each specification are crossed with nest indicators. Conditioning on the correct nest structure eliminates ambiguity about the source of any rejection: it must reflect market misspecification.

A.5.3 Known Markets, Unknown Nests

Suppose instead that the market definition m is well-established but the nest structure is uncertain. This case has no analogue in the plain logit model, where no nest structure exists, and it is where the observable-characteristics benchmark of Section A.4 is most useful.

Let the fine benchmark be $(h \times m)$ fixed effects, where h is either the full interaction X^d_{full} or a grouping known to be finer than all candidates under consideration. The candidate uses $(\hat{g} \times m)$ fixed effects for a proposed nesting schedule $\hat{g} = \hat{H}(X^d)$:

$$\hat{\theta}^{\text{fine}}: \quad (h \times m) \text{ fixed effects,} \tag{35}$$

$$\hat{\theta}^{\text{cand}}: \quad (\hat{g} \times m) \text{ fixed effects.} \tag{36}$$

Under H_0, $\hat{g}$ is at least as fine as the true nest g, so $(\hat{g} \times m)$ absorbs Λ_{gm} and both estimators are consistent. Under H_1, $\hat{g}$ is too coarse: some $(\hat{g} \times m)$ cells pool products across true nests, and the nest-level component $[\sigma/(1-\sigma)]\ln(s_{gm})$ in equation (28) is not absorbed.

When nests are functions of observables, the fine benchmark is always available: the full interaction X^d_{full} refines every candidate $\hat{H}(X^d)$ by construction. This allows the researcher to test competing nesting schedules—for example, {nonstop, connecting} versus {nonstop-legacy, nonstop-LCC, connecting}—against a common benchmark without needing to specify the true nest assignment in advance.

The practical constraint is that the number of products per (X^d_{full}, m) cell must be large enough to identify θ_α with reasonable precision. When the full interaction produces thin cells, the researcher should use the coarsest grouping that is still known to refine all candidates under consideration, trading universality for statistical power.

A.6 Summary of Test Structure

Table 6 summarizes the three cases.

Table 6 – PW Tests in the Nested Logit Framework

Scenario	Fine FE	Candidate FE	Object tested
Simultaneous	$h \times s$	$\hat{g} \times \hat{m}$	Joint adequacy of nests and markets
Known nests	$g \times s$	$g \times \hat{m}$	Market definition, given nests
Known markets	$h \times m$	$\hat{g} \times m$	Nest definition, given markets

Note: In all three cases, the parameter under test is $\theta_\alpha = \alpha/(1-\sigma)$, and the PW-2SLS test is implemented by differencing influence functions across the two fixed-effects specifications. $h \subset g$ denotes subnests (or the full observable interaction X^d_{full}), $s \subset m$ denotes submarkets.

A.7 Caveats

Four caveats are specific to the nested logit extension; the logit-level caveats discussed in Sections 3 and 4 continue to apply.

Non-separability of α and σ. The direct regression identifies only the composite $\theta_\alpha = \alpha/(1-\sigma)$, not the structural parameters individually. The test therefore cannot distinguish between, for example, $(\alpha = -3, \sigma = 0.5)$ and $(\alpha = -1.5, \sigma = 0)$, which both yield $\theta_\alpha = -3$. Separate identification of α and σ requires the structural specification with correctly specified nests and markets. Accordingly, the PW test disciplines whether a candidate (g, m) pair is adequate for θ_α, not whether it recovers α and σ individually.

Heterogeneous nesting parameters. The derivation in Section A.2 assumes σ is constant across nests. If σ varies by nest ($\sigma = \sigma_g$), then dividing equation (25) by $(1-\sigma_g)$ introduces a nest-varying coefficient on p and X. In that case, θ_α is not a single parameter and the direct regression (27) is misspecified even with correct (g, m) fixed effects. The testing framework requires either a constant σ or a restriction to within-nest-type comparisons where σ is plausibly homogeneous.

Nests must be functions of observables. The universal fine benchmark of Section A.4 requires that the true nest assignment satisfy $g_j = H(X^d_j)$ for observable categorical characteristics X^d. If nests depend on unobserved product attributes—for example, if consumer substitution patterns are driven by latent quality dimensions not captured by X^d—then the full interaction of observables need not refine the true nest structure, and the fine benchmark may itself be inconsistent. In such cases, the test remains valid when the researcher can identify a subnest grouping $h \subset g$ on other grounds, but the observable-characteristics shortcut does not apply.

Degrees of freedom. Fixed-effects cells multiply across two dimensions: the number of $(h \times s)$ cells is the product of the number of subnest and submarket groups. If the number of products per cell is small, the fine benchmark may consume a large share of the within-cell variation, reducing precision and potentially undermining the power of the PW test. The tradeoff between consistency of the fine benchmark and estimation precision is sharper in the nested logit setting than in the single-dimension logit case, and is especially acute when h is the full observable interaction.

B Random-Coefficients Logit Simulation

Modern demand applications often allow richer substitution patterns than aggregate logit. The exact fixed-effects identification result in the paper is specific to aggregate logit and nested logit, so the random-coefficients exercise is best interpreted as a robustness and external-validity check rather than as a new theorem. Accordingly, this appendix asks a narrower question: when data are generated by a BLP-style demand system with endogenous prices and true county markets, does the PW diagnostic still tend to flag candidate aggregations that change the price coefficient implied by a simpler logit-style estimating equation?

I simulate 50 states, 10 counties per state, 5 tracts per county, and roughly 15 products per true county market for 250 replications. Demand is generated with a random coefficient on x_1, product-level demand shocks, a county-level demand component, and endogenous Bertrand-Nash prices solved with `pyblp` using the true county market definition. I then estimate misspecified logit-style equations: structural logit under the true county shares and a too-coarse state aggregation, plus direct log-quantity IV regressions with state, county, and tract fixed effects.

The coefficient estimates in Table 7 should therefore not be read as recovering the true BLP mean price coefficient. They summarize the price coefficient implied by the simpler logit-style estimators under alternative market definitions. The PW comparisons in Table 8 then test whether those implied coefficients are asymptotically equivalent to the tract benchmark.

The results support the more modest diagnostic interpretation. With a clean instrument, all five misspecified logit-style estimates cluster tightly near -1.40. The associated PW rejection rates stay close to nominal size, with tract-versus-structural-state rejecting between 5-6% of replications. When the instrument inherits county-level dependence, the pattern changes sharply. The Direct-County and tract estimators remain near -1.40, while the structural county, structural state, and Direct-State coefficients shift to -0.79, -0.78, and -0.81, respectively. Correspondingly, tract-versus-structural-state rejects in 100% of replications and tract-versus-state-direct rejects in 100%, whereas tract-versus-county-direct stays near nominal size at 6%.

Overall, the results suggest that the PW diagnostic can remain informative as a specification diagnostic even when the exact aggregate-logit algebra is no longer available. What it does not show is that the simpler logit estimators are correctly specified for the random-coefficients DGP, nor that non-rejection should be interpreted as structural validation.

Table 7 – MC-BLP Coefficients

Panel A: Structural logit estimators		
Estimator	Clean instrument (1)	Contaminated instrument (2)
Structural true county	−1.40	−0.79
	(0.04)	(0.03)
Structural too-coarse state	−1.40	−0.78
	(0.04)	(0.03)

Panel B: Direct logit estimators		
Estimator	Clean instrument (1)	Contaminated instrument (2)
Direct-State	−1.40	−0.81
	(0.04)	(0.03)
Direct-County	−1.40	−1.40
	(0.01)	(0.01)
Direct-Tract	−1.40	−1.40
	(0.01)	(0.01)

Note: Each first line reports Monte Carlo means for the price coefficient from a misspecified logit-style estimating equation, even though the data are generated by a random-coefficients logit DGP. The line below reports Monte Carlo standard deviations in parentheses. True pyblp markets are counties in every replication; the structural estimators differ only in how the shares and outside shares are reconstructed for estimation.

Table 8 – MC-BLP PW Tests

Panel A: Clean instrument		
Comparison	Mean t-statistic (1)	Rejection rate (2)
Tract vs Structural true county	0.02	0.06
Tract vs Structural too-coarse state	−0.05	0.06
Tract vs County direct	0.01	0.05
Tract vs State direct	−0.05	0.05
County vs State direct	−0.05	0.06
Panel B: Contaminated instrument		
Comparison	Mean t-statistic (1)	Rejection rate (2)
Tract vs Structural true county	−21.34	1.00
Tract vs Structural too-coarse state	−21.40	1.00
Tract vs County direct	−0.01	0.06
Tract vs State direct	−20.10	1.00
County vs State direct	−20.85	1.00

Note: Each row reports the mean PW t-statistic and the Monte Carlo rejection rate for a two-sided 5% test. The tract estimator is the fine benchmark in comparisons involving the structural, county direct, and state direct estimators. High rejection rates should be interpreted as evidence that the coarser or structurally reconstructed specification implies a materially different misspecified logit coefficient.

C Additional Details for the Airline Application

This appendix collects implementation details for the airline application in Section 5. The goal is to keep the main text focused on the market-definition test itself while documenting the construction of the estimation sample, the candidate market definitions, the estimating equations, and the instrument sets in a single place.

C.1 Data and Sample Construction

The application uses the replication data from Aryal et al. (2024), based on the DOT DB1B Airline Origin and Destination Survey. The DB1B is a 10% sample of airline tickets, and the analysis uses the second quarter of each year from 1998 to 2016. Following Berry and Jia (2010) and Berry et al. (2006), passenger counts are multiplied by ten to recover estimated totals.

The raw sample contains 1,123,368 carrier-market-year observations for 16 ticketing carriers. Market size is defined as

$$M_m = 0.9 \times \sqrt{\text{pop}_o \times \text{pop}_d},$$

where pop_o and pop_d are 2010 Census CBSA populations for the origin and destination markets. The outside share is computed as

$$s_{0m} = 1 - \sum_j s_{jm}$$

at each candidate market definition.

The estimation sample imposes the same restrictions across market definitions. I drop thin markets with fewer than 200 estimated passengers, infrequent carriers, observations with outside share outside the interval $[0.5, 0.999]$, and singleton observations. The resulting common sample used for the direct IV and PW comparisons contains 863,182 observations across 18,982 airport pairs and 16 carriers. The median outside share is 0.987, close to the value reported by Berry and Jia (2010).

C.2 Candidate Market Definitions

The application compares three candidate market definitions arranged in a hierarchy.

1. **Airport pair (fine):** a directional airport-to-airport market, such as DCA–ORD.
2. **Origin-airport × destination-city (intermediate):** the origin airport is held fixed while destination airports within the destination MSA are pooled, so that DCA–Chicago combines ORD and MDW.
3. **City pair (coarse):** the origin MSA and destination MSA are each pooled, as in Washington–Chicago.

Note, this hierarchy is not *strict* because some cities only have one airport, and so in these cases the airport and city aspects of the market are identical.

Each market definition is indexed separately by year. Airport-to-MSA assignments are based on a static IATA-to-CBSA crosswalk covering the 170 airports in the estimation sample. By construction, each finer definition is nested (though not necessarily strictly) within the next coarser definition, which is the hierarchical structure required by the PW procedure.

C.3 Estimating Equations

The main text reports results from two estimators.

Direct estimator. The direct IV estimator is

$$\ln[q_{jmt}] = \alpha(p_{jmt}/y_{ot}) + \beta X_{jmt} + \mu_j + \lambda_{mt} + u_{jmt}, \tag{37}$$

where q_{jmt} is carrier passenger quantity, p_{jmt}/y_{ot} is the price-income ratio, X_{jmt} is the vector of included controls, μ_j are carrier fixed effects, and λ_{mt} are market-by-year fixed effects. Under this approach, the market definition enters only through the fixed effects.

Structural logit estimator. The structural logit estimator is

$$\ln[s_{jmt}] - \ln[s_{0mt}] = \alpha(p_{jmt}/y_{ot}) + \beta X_{jmt} + \mu_j + \lambda_m + \lambda_t + \xi_{jmt}, \tag{38}$$

where product and outside shares are recomputed at each candidate market definition. Under this approach, the market definition affects both the dependent variable and the fixed effects structure.

The endogenous regressor in both specifications is

$$\text{price\_income}_{jmt} = \frac{p_{jmt}}{y_{ot}/1000}, \tag{39}$$

where p_{jmt} is the carrier's average fare and y_{ot} is origin-market median household income. This scaling gives α the interpretation of a marginal utility of income.

C.4 Included Controls

The included controls are

$$X_{jmt} = \left(\text{NonstopPassengerShare}_{jmt},\ \text{MeanCircuity}_{jmt}\right). \tag{40}$$

Nonstop passenger share. This variable is the fraction of a carrier's passengers in a carrier-market-year cell who travel on nonstop itineraries. At the itinerary level, an itinerary is classified as nonstop when no second flight segment is observed. The carrier-market-year measure is then constructed as the passenger-weighted share of traffic that is nonstop.

Mean circuity. Mean circuity is the passenger-weighted average of itinerary circuity, where

$$\text{circuity} = \frac{\text{total itinerary distance}}{\text{nonstop market distance}}. \tag{41}$$

A value of one corresponds to a nonstop itinerary, while larger values indicate more circuitous routing.

C.5 Excluded Instrument Sets

The empirical application reports results for three excluded-instrument configurations.

C.5.1 Own Cost Shifters

The own cost-shifter instrument set is

$$Z_{jmt}^{\text{cost}} = \left(\text{HasNonstop}_{jmt}, \text{MeanOwnedRegional}_{jmt}, \text{MeanUnownedRegional}_{jmt}\right). \tag{42}$$

HasNonstop. This indicator equals one if the carrier offers at least one nonstop itinerary in the market-year.

MeanOwnedRegional and MeanUnownedRegional. These variables are passenger-weighted averages of itinerary-level regional subcontracting measures and summarize the extent to which the carrier's service on the route is operated by owned regional affiliates or by unowned regional partners.

C.5.2 BLP-Style Rival Characteristics

The BLP-style rival instrument set is

$$\begin{aligned} Z_{jmt}^{\text{BLP}} = (&\text{RivalNonstopCount}_{jmt}, \text{RivalPctNonstop}_{jmt}, \text{RivalMeanCircuity}_{jmt}, \\ &\text{RivalMeanOwnedReg}_{jmt}, \text{RivalMeanUnownedReg}_{jmt}, \text{NumRivals}_{mt}). \end{aligned} \tag{43}$$

These variables are constructed as leave-own-out market aggregates. For each carrier in a market-year, the rival instruments summarize the characteristics of competing carriers in the same market-year, excluding the carrier's own observation. Rival nonstop count is the number of rival carriers that offer at least one nonstop itinerary. Rival nonstop share is the leave-own-out average nonstop passenger share among rivals. Rival circuity and rival regional usage are the corresponding leave-own-out averages of service characteristics. Number of rivals is the number of carriers present in the market-year.

C.5.3 Combined Cost + BLP Set

The third instrument set stacks the own cost shifters and BLP-style rival characteristics together.

C.6 Aggregation Across Market Definitions

For the direct IV specifications, the same control and instrument definitions are used across the three candidate market definitions, with the market definition entering only through the fixed effects.

For the structural logit specifications, the data are re-aggregated to each candidate market definition before estimation. When this occurs, carrier-level variables such as fares, nonstop passenger share, circuity, regional usage, and network size are aggregated using passenger-weighted means, while `has_nonstop` is updated as the maximum across the underlying finer routes. The rival-characteristic instruments are then recomputed at the relevant market definition using leave-own-out formulas.

C.7 First-Stage Results

Table 9 reports the first-stage Kleibergen–Paap rk Wald F statistics for each instrument set under each market definition and estimation approach.

The first stages are strong in all cases. Accordingly, the unusual second-stage behavior of the BLP-rival and combined instrument sets in the structural logit does not appear to be driven by weak identification.

C.8 Additional Discussion of the Second-Stage Results

The main text emphasizes the broad patterns in Table 3. A few additional remarks are useful.

First, the own cost-shifter IVs yield stable negative coefficients across candidate market definitions in both the direct and structural logit specifications. This stability is consistent with the view that these instruments primarily capture variation that is less sensitive to the precise level of market aggregation.

Second, the BLP-rival and combined instrument sets behave differently across the two estimators. In the direct specification, the estimates become more negative as the market definition becomes coarser. This is exactly the kind of pattern the PW test is designed to detect, since coarser fixed effects may leave residual market-level demand variation in the

Table 9 – First-Stage Kleibergen–Paap rk Wald F Statistics

Market definition	Airport Pair (1)	Orig.Airport–Dest.City (2)	City Pair (3)
Panel A: Direct, Interacted market-by-year FE			
Own cost shifters	207.8	155.6	159.9
BLP rival chars	129.6	57.3	65.9
Cost + BLP	137.4	90.4	107.5
Market × year FE	Yes	Yes	Yes
Carrier FE	Yes	Yes	Yes
N	863,182	863,182	863,182
Panel B: Structural Logit, Separate market + year FEs			
Own cost shifters	151.1	137.4	125.1
BLP rival chars	291.7	281.2	273.1
Cost + BLP	241.2	231.2	221.8
Market FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Carrier FE	Yes	Yes	Yes
N	863,209	794,987	730,308

Note: This table reports first-stage Kleibergen–Paap rk Wald F statistics from regressions of the price/income ratio on the excluded instruments. Panel A uses the direct specification with interacted market-by-year fixed effects. Panel B uses the structural logit specification with separate market and year fixed effects. All specifications include carrier fixed effects and the two carrier-route-year controls.

structural error.

Third, in the structural logit specification, the BLP-rival and combined instrument sets generate positive coefficients despite strong first stages. The natural interpretation is that, in this implementation, the rival-characteristic instruments still contain demand-side variation when the specification uses only separate market and year effects rather than interacted market-by-year effects. That pattern is informative about instrument validity under alternative fixed-effect structures, even if the structural estimates themselves are not economically plausible.

C.9 Robustness: Carrying Forward Fine-Level Rival IVs

The baseline structural logit specification recomputes the leave-own-out rival instruments separately at each candidate market definition after aggregation. An alternative approach is to hold the airport-pair-level rival IVs fixed and aggregate them upward to coarser definitions using the same passenger-weighted averaging used for own characteristics. The two approaches are identical at the airport-pair level by construction.

Table 10 compares these two approaches. The coefficient estimates and first-stage statistics are very similar across them, with only minor differences at the coarser market definitions. This robustness check indicates that the structural-logit results are not sensitive to whether the rival IVs are recomputed after aggregation or instead carried forward from the fine market level.

C.10 Interpretation

The included controls are intended to capture observed service quality and routing characteristics, while the excluded instruments provide variation intended to shift the endogenous price-income ratio. Because the application reports results across several instrument sets and across both direct and structural specifications, the appendix lists all definitions explicitly even though the main text emphasizes only the patterns most relevant for the market-definition test.

Table 10 – Structural Logit: BLP Rival IV Comparison

	Airport Pair (1)	Orig.Airport–Dest.City (2)	City Pair (3)
BLP rival (recomputed)	1.02	1.03	0.92
	(0.03)	(0.03)	(0.03)
BLP rival (airport-pair)	1.02	1.03	0.98
	(0.03)	(0.03)	(0.04)
First-stage F (recomputed)	291.7	281.2	273.1
First-stage F (airport-pair)	291.7	251.3	232.9
Market FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Carrier FE	Yes	Yes	Yes
N	863,209	794,987	730,308

Note: "Recomputed" recomputes leave-own-out rival IVs at each candidate market level after aggregation. "Airport-pair" carries the airport-pair-level rival IVs forward to coarser definitions by passenger-weighted averaging. Both approaches recompute shares at the candidate market level.

D Bank Markets Application

Market definition has long been a contested issue in empirical banking research. Traditional geographic markets such as counties, metropolitan statistical areas (MSAs), or states are often adopted for institutional or regulatory convenience, yet consumer substitution patterns in deposit markets may operate over narrower or broader geographic areas. Moreover, banks typically operate multi-branch networks, and deposit pricing decisions are often set at the bank or regional level rather than at the individual branch level.

This section applies the Papke–Wooldridge (PW) test developed above to branch-level banking data. The goal is to assess whether commonly used geographic market definitions provide a sufficiently rich fixed-effect structure to satisfy the identifying assumptions of a branch-level demand model with endogenous bank-level pricing.

We will consider three sets of PW tests: tracts versus counties, tracts versus metropolitan statistical areas (MSAs), and tracts versus commuting zones (CZs). Our submarket will be census tracts. Both MSAs and CZs nest counties; however, MSAs and CZs are not coterminous.

D.1 Branch Demand Model

The underlying demand model assumes that depositors choose among bank branches. The empirical question is which geographic market best captures that choice set.

Branch-level demand is modeled using a log-quantity specification motivated by the aggregate logit framework developed in Section 2.2. Let b index branches, $j(b)$ index the parent bank of branch b, $m(b)$ index the geographic market in which branch b is located, and t index calendar years.

The structural logit function is:

$$\log(s_{bjmt}) - \log(s_{0,mt}) = \alpha\, p_{bjmt} + \beta X_{bjmt} + \varepsilon_{bjmt}, \tag{44}$$

where s_{bjmt} is the branch market share and $s_{0,mt}$ is the outside good share, p_{bjmt} is a deposit rate (paid to the depositor), X_{bjmt} is a vector of branch characteristics, and ε_{bjmt} is an unobserved branch-level demand shock. The branch market shares are measured based on the volume of deposits at a given branch.

D.2 Data

To estimate the model, I use data spanning 2015 to 2025. For branch data, I use the Summary of Deposits, which is annual regulatory data on the location and deposits of all bank branches, plus limited branch characteristics. For bank data, I use Call Reports, which is quarterly regulatory data on bank financial data. I augment the banking data with quarterly averages of the 1-Year Treasury rates, which we use to measure funding shocks.

D.3 Branch Variables

We do not observe branch level pricing, so we instead assume that deposit rates are set at the bank level. We discuss this variable more below.

The branch characteristics include branch age, an indicator for main-office branches, and an indicator for full-service branches. These variables capture persistent differences in branch size, functionality, and visibility that are plausibly relevant for deposit demand.

I use bank fixed effects to absorb time-invariant bank attributes such as brand reputation, long-run service quality, and persistent differences in product menus.

Bank-Level Price Measurement Branch-level deposit rates are not observed. We construct bank-level average deposit rates from Call Report data and assign the rate to all branches of the bank in a given year. Recent work supports that banks appear to set rates nationally or regionally rather than locally (Granja and Paixao, 2023; Begenau and Stafford, 2023).

Specifically, let q index calendar quarters. The baseline deposit rate measure is defined as an annualized flow-stock ratio over the four quarters ending in Q2 of year t:

$$p_{jt} = \frac{\sum_{q=\mathrm{Q3}(t-1)}^{\mathrm{Q2}(t)} \mathrm{DepositInterestExpense}_{jq}}{\frac{1}{4}\sum_{q=\mathrm{Q3}(t-1)}^{\mathrm{Q2}(t)} \mathrm{Deposits}_{jq}}. \tag{45}$$

This measure captures the average interest rate paid on deposits over the period most relevant for the Q2 deposit stock. Any residual within-bank heterogeneity in branch-level pricing is absorbed into the error term.

D.4 Endogeneity and Instrumental Variables

Bank-level deposit rates are potentially endogenous if unobserved changes in deposit demand are correlated with pricing decisions. To address this concern, equation (44) is estimated by two-stage least squares (2SLS) using instruments constructed from bank funding structure and national interest rate movements.

Our instrument is a funding shock exposure. This instrument type follows a shift-share design that exploits heterogeneity in banks' liability composition combined with common national rate shocks.

Let Δr_t denote the change in a national short-term interest rate between Q2 of year $t-1$ and Q2 of year t. Let exposure measures be defined using lagged balance sheet shares measured in Q2 of year $t-1$: (1) wholesale funding share, (2) brokered deposit share, and (3) large time deposit share.

The baseline instruments are

$$Z_{bt}^{(k)} = \text{Exposure}_{b,t-1}^{(k)} \times \Delta r_t, \qquad k = 1, 2, \ldots \tag{46}$$

where exposures vary across banks and Δr_t is common to all banks.

These instruments shift banks' marginal cost of funds and therefore affect deposit pricing, while their interaction structure ensures that identification comes from cross-bank differences rather than aggregate time variation.

D.5 Empirical Specification

The PW test is applied at the branch level, with census tracts as the submarket definition. We compare three candidate geographic markets against tracts: counties, commuting zones (CZs), and metropolitan statistical areas (MSAs). For the MSA comparison, areas that do not belong to any MSA are assigned to their state, so that all observations are retained and the sample does not shrink relative to the county or CZ comparisons.

The estimated demand equation follows equation (44), with branch-by-year observations. The dependent variable is log branch deposits. The endogenous variable is the bank-level deposit rate constructed in equation (45). Branch controls include an indicator for full-service branches, an indicator for main-office branches, the log of branch age in months, and the log of months since the branch was most recently purchased. Bank fixed effects are included to absorb time-invariant bank-level attributes; this implies that banks operating a single branch are excluded from estimation. Area fixed effects are specified as geography-by-year interactions at the candidate market level, absorbing local time-varying economic conditions within each candidate market.

The deposit rate is instrumented using the funding-shock exposure instruments described in equation (46), constructed from lagged brokered deposit and wholesale funding shares interacted with changes in the national short-term rate.

We present results under two sample definitions. In the first version, each pairwise PW test (tracts vs. counties, tracts vs. CZs, tracts vs. MSAs) is estimated on its own unrestricted sample; that is, we do not require observations to be common across the three comparisons. In the second version, we restrict to a common sample across all three tests.

D.6 Results

Table 11 presents the 2SLS estimates under all seven specifications and Table 12 presents the corresponding PW tests. The two sample versions yield sharply different conclusions, and understanding the source of the discrepancy is central to interpreting the results.

Instrument strength is not a concern in any specification. Cluster-robust first-stage F-statistics range from 66.2 to 72.7 across all seven regressions, well above conventional thresholds, and are stable across both sample definitions.

Consider first the unrestricted sample, in which each pairwise comparison is estimated on its own sample. The PW test rejects all three candidate markets with overwhelming precision ($p < 0.001$ in each case), and the estimated price coefficient under tract-by-year fixed effects, 26.85, is noticeably smaller than the estimates under county, CZ, and MSA fixed effects, which range from 29.09 to 30.50.[5] However, this pattern is an artifact of a compositional difference across specifications. The tract specification contains 717,475 observations, while the county, CZ, and MSA specifications each contain roughly 886,000—a gap of nearly 170,000 branches. The missing observations are branches located in census tracts that contain only a single branch. In the tract specification, these single-branch tracts are perfectly absorbed by the tract-by-year fixed effects and contribute no identifying variation to the price coefficient; in the coarser-geography specifications, they remain in the sample and help identify the estimate. The PW test therefore compares a price estimate identified from multi-branch tracts against estimates identified from a broader population of branches, and the resulting differences reflect this compositional mismatch rather than a failure of the coarser market definitions.

The common-sample results resolve the discrepancy. Restricting all four specifications to the same 717,475 observations eliminates the compositional difference, and the PW test fails to reject the null for all three candidate markets: the p-values are 1.000 for counties, 0.997 for commuting zones, and 0.997 for MSAs. The estimated price coefficient under tract-by-year fixed effects remains 26.85, while the estimates under county, CZ, and

[5] Since deposit rates are measured in decimals, a 100bp increase (0.01) implies roughly a 0.27-0.31 increase in log deposits.

MSA fixed effects are 26.08, 29.03, and 30.13, respectively. Although these point estimates still differ, the PW test finds the differences statistically indistinguishable from zero. Conditional on the exogenous variation identified by the funding-shock instruments, tract-level deposit demand heterogeneity does not alter the estimated price coefficient relative to what is obtained under any of the three coarser geographic market definitions.

Tract fixed effects clearly capture meaningful variation in deposit levels, most of it likely reflecting persistent differences in tract-level demographics and branch density. The PW test, however, isolates whether this additional variation changes the identified relationship between deposit rates and deposit demand. It does not, and the failure to reject on the common sample indicates that county, CZ, and MSA fixed effects provide a sufficiently rich structure for the purposes of estimating the price coefficient in this demand model.

Table 11 – Branch-Level Deposit Demand: 2SLS Estimates

		Unrestricted sample			Common sample		
	Tract	County	CZ	MSA	County	CZ	MSA
Deposit rate	26.85	29.09	30.10	30.50	26.08	29.03	30.13
	(6.41)	(5.12)	(4.93)	(4.92)	(5.73)	(5.53)	(5.51)
Other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year × Geo. FE	Tract	County	CZ	MSA	County	CZ	MSA
First-stage F	72.65	66.17	67.36	69.46	66.20	66.36	67.97
Observations	717,475	886,380	887,589	887,649	717,475	717,383	717,383

Note: Dependent variable is log branch deposits (in millions). Deposit rate is instrumented using lagged brokered deposit and wholesale funding shares interacted with the change in the federal funds rate. Bank-clustered standard errors in parentheses. First-stage F-statistics are cluster-robust. The tract specification is identical across sample versions.

Table 12 – Papke–Wooldridge Market Definition Tests

Comparison	$\hat{\alpha}_{\text{tract}}$	$\hat{\alpha}_{\text{coarse}}$	Difference	t-statistic	p-value
Panel A: Unrestricted sample					
Tract vs. County	26.85	29.09	−2.24	−17.60	< 0.001
Tract vs. CZ	26.85	30.10	−3.25	−17.40	< 0.001
Tract vs. MSA	26.85	30.50	−3.65	−17.40	< 0.001
Panel B: Common sample					
Tract vs. County	26.85	26.08	0.77	0.00	1.000
Tract vs. CZ	26.85	29.03	−2.18	0.00	0.997
Tract vs. MSA	26.85	30.13	−3.28	0.00	0.997

Note: The PW test assesses whether tract-by-year fixed effects provide statistically significant improvement over the coarser geography-by-year fixed effects in the instrumented demand equation. $\hat{\alpha}_{\text{tract}}$ and $\hat{\alpha}_{\text{coarse}}$ are the 2SLS estimates of the deposit rate coefficient under tract-by-year and coarser-geography-by-year fixed effects, respectively. The null hypothesis is that the two coefficients are equal. The unrestricted sample allows each comparison to use its own sample; the common sample restricts all specifications to branches in tracts with more than one branch. p-values are one-sided.